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**Ministry of Education**

**Qassim University**

**Applied Sciences College**

**Department of Cyber Security**



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وزارة التعليم

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**Cyber security program**

**Graduation Project**

**CYS\*\*\*\*– Term \*\*\***

# **Intrusion Detection Systems (IDS) & Intrusion Protection Systems (IPS)**

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**PROJECT TITLE**

**Intrusion Detection  
Systems (IDS) &  
Intrusion Protection  
Systems (IPS)**

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# TABLE OF CONTENTS

|                                                                |           |
|----------------------------------------------------------------|-----------|
| ACKNOWLEDGMENT .....                                           | 3         |
| Table of Contents .....                                        | 4         |
| Chapter One .....                                              | 7         |
| 1.1 What is IDS? .....                                         | 7         |
| 1.2 What is IPS? .....                                         | 8         |
| 1.3 What is the Difference between IDS and IPS? .....          | 10        |
| 1.4 IDS & IPS Working Together .....                           | 12        |
| 1.5 How Does IDS/IPS Detect Threats?.....                      | 12        |
| 1.6 Which is better, IDS or IPS? .....                         | 12        |
| 1.7 Can an IPS prevent a DDoS attack? .....                    | 13        |
| 1.8 Open Source IDS and IPS Tools.....                         | 13        |
| Chapter Two.....                                               | 17        |
| <b>install Ubuntu 20.04 Desktop On VMware Workstation.....</b> | <b>17</b> |
| <b>Prerequisite: .....</b>                                     | <b>17</b> |
| 2.1 Create the Ubuntu 20.04 virtual machine. ....              | 17        |
| 2.1.1 Choose the hardware compatibility. ....                  | 18        |
| 2.1.2 Attach the ISO image.....                                | 18        |
| 2.1.3 Set up the user credential for ubuntu 20.04. ....        | 19        |
| 2.1.4 Name the VM and the location. ....                       | 20        |
| 2.1.5 Configure the CPU.....                                   | 20        |
| 2.1.6 Configure the Memory.....                                | 21        |
| 2.1.7 Configure the networking. ....                           | 22        |
| 2.1.9 Create a virtual hard disk. ....                         | 24        |
| 2.1.10 Define the storage size. ....                           | 24        |
| 2.2 Installing Empty Copy of Operating System:.....            | 27        |
| <b>2.3 Installing SNORT .....</b>                              | <b>28</b> |
| <b>2.3.1 Update: Virtual-machine: .....</b>                    | <b>28</b> |

|                                                                |    |
|----------------------------------------------------------------|----|
| 2.3.2 install build tools for snort:.....                      | 28 |
| 2.3.3 create directory where kept source files for snort:..... | 29 |
| 2.3.4 Open directory:.....                                     | 30 |
| 2.3.5 Install lipDAQ:.....                                     | 30 |
| 2.3.6 Open directory:.....                                     | 31 |
| 2.3.7 Install configuration: .....                             | 31 |
| 2.3.8 Make:.....                                               | 32 |
| 2.3.9 Make install: .....                                      | 32 |
| 2.3.11 install snort3: .....                                   | 33 |
| 2.3.12 Open files snort3: .....                                | 34 |
| 2.3.13 Open directory:.....                                    | 34 |
| 2.3.14 install configure and make direcotry: .....             | 34 |
| 2.3.15 Build directory:.....                                   | 35 |
| 2.3.16 Make:.....                                              | 35 |
| 2.3.17 Make install: .....                                     | 36 |
| 2.3.18 Verify Snort 3 Installation: .....                      | 36 |
| References.....                                                | 38 |

# CHAPTER ONE

# CHAPTER ONE

## Introduction

### 1.1 What is IDS?

IDS or “intrusion detection systems” is a specially made software designed to protect a network or system against malicious traffic. Any dangerous looking activity is often reported to an administrator. An intrusion detection system monitors traffic on your network, analyzes that traffic for signatures matching known attacks, and when something suspicious happens, you're alerted. In the meantime, the traffic keeps flowing.

#### Intrusion Detection System (IDS)

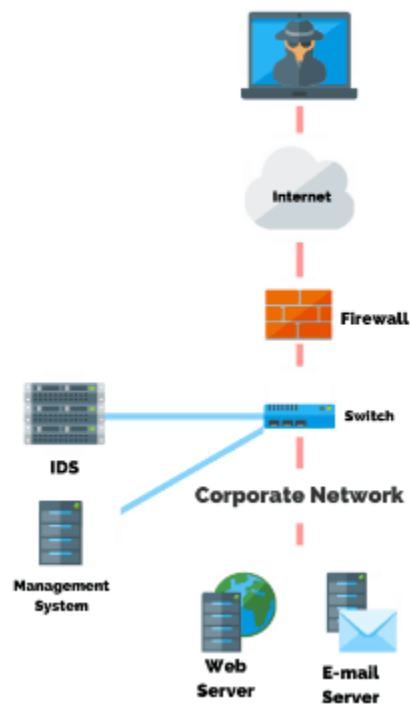


Figure 1. IDS

Five main types of IDS exist.

1. **Network:** Choose a point on your network and examine all traffic on all devices from that point.
2. **Host:** Examine traffic to and from independent devices within your network, and leave all other devices alone.

3. **Protocol-based:** Place protection between a device and the server, and monitor all traffic that goes between them.
4. **Application protocol-based:** Place protection within a group of servers and watch how they communicate with one another.
5. **Hybrid:** Combine some of the approaches listed above into a system made just for you.

No matter what type of IDS you choose, the underlying function is similar. You'll use passive technology to detect an intrusion. When something is found, you'll get an alert about it. The system might detect problems with:

- **Patterns.** The technology flags unusual requests, very large packet sizes, or anything that seems out of the ordinary for your system at that time.
- **Prior attacks.** The technology flags anything on your server that's been used in a known and successful attack on another server.
- **Machine learning.** The system picks up information about everything that happens on your server in an average day, and that data helps to refine the protections in place.

A system like this comes with plenty of advantages. You can set one up very quickly, and it's hard for attackers to spot your protections. But by the time you're notified of an attack in progress, you may have lost valuable time you could use to protect your assets.

## 1.2 What is IPS?

IPS or “intrusion prevention system is an application that works by identifying, reporting, and even preventing potential malware. An intrusion prevention system also monitors traffic. But when something unusual happens, the traffic stops altogether until you investigate and decide to open the floodgates again. The goal of an IPS is to prevent damage. While you're kept in the loop about the attack, the system is already working to keep things safe. An IPS can protect against exterior intruders. But people within the organization can also take steps that harm your security. An IPS can protect against these actions too, so it can help train the employees about what is allowed and what is not.

There are multiple types of Intrusion Protection Systems:

1. **Network:** Analyze and protect traffic on your network.



2. **Wireless:** Observe anything happening within a wireless network and defend against an attack launched from there.
3. **Network behavior:** Spot attacks that involve unusual traffic on your network.
4. **Host-based:** Scan events that occur within a host you specify.

### Intrusion Prevention System (IPS)

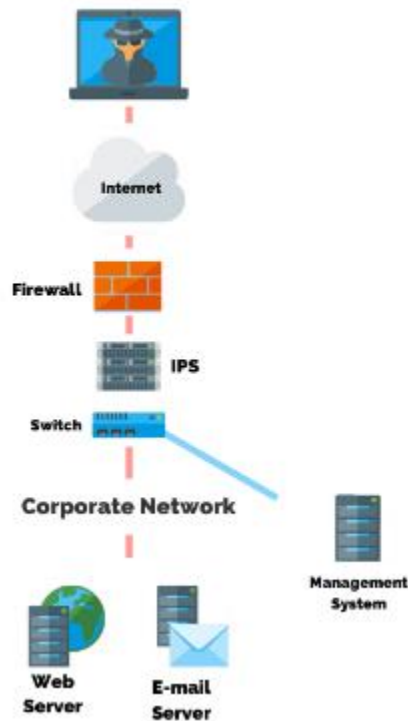


Figure 2. IPS

Most administrators install an IPS right behind the firewall. But it's clear that you can set up different configurations to protect your company's assets in a much customized way. No matter what type you choose, the IPS will take a predictable set of steps. An IPS monitors your network for suspicious activity or attacks in progress. When an anomaly is spotted, you're alerted. But in the interim, the system springs to action. It might:

- Close sessions. It can identify the entry point for the unusual activity and block it from continuing. That might mean terminating the TCP session, blocking an IP address, or some similar step.
- Strengthen firewalls. The system might identify the gap in the firewall that allowed the attack to enter. A change in programming could prevent something similar in the future.

- **Clean up.** The system can scan for damaged or malicious content within the server and remove it.

The IPS works in real time, and the technology inspects every packet moving through the protected space. It can take swift and significant action in response to an attack, and while that can keep you protected, these systems can throw up false positives. When that happens, traffic can slow for no reason whatsoever.

At the end of an attack and cleanup, you have a report of what went wrong and what the system did to fix it. You can use this data to help you strengthen your system against a future attack. And you can walk back through the protections the system made and undo them if you feel they aren't worthwhile.

### 1.3 What is the Difference between IDS and IPS?

Similar to a firewall, IPS is deployed inline to the traffic flow. IPS is an active network component that examines every passing packet and takes the correct remedial action per its configuration and policy. In contrast, IDS is a passive component typically not deployed inline and instead monitors the traffic flow via span or tap technology to then raise notifications. As a result, we can conclude the similarities and differences between IDS and IPS as follow:

#### **Both systems can:**

- **Monitor.** After setup, these programs can look over traffic within parameters you specify, and they will work until you turn them off.
- **Alert.** Both programs will send a notification to those you specify when a problem has been spotted.
- **Learn.** Both can use machine learning to understand patterns and emerging threats.
- **Log.** Both will keep records of attacks and responses, so you can adjust your protections accordingly.

#### **But they differ due to:**

- **Response.** An IDS is passive, while an IPS is an active control system. You must take action after an IDS alerts you, as your system is still under attack.

- **Protection.** Arguably, an IDS offers less help when you're under threat. You must figure out what to do, when to do it, and how to clean up the mess. An IPS does all of this for you.
- **False positives.** If an IDS gives you an alert about something that isn't troublesome at all, you're the only one inconvenienced. If an IPS shuts down traffic, many people could be impacted.

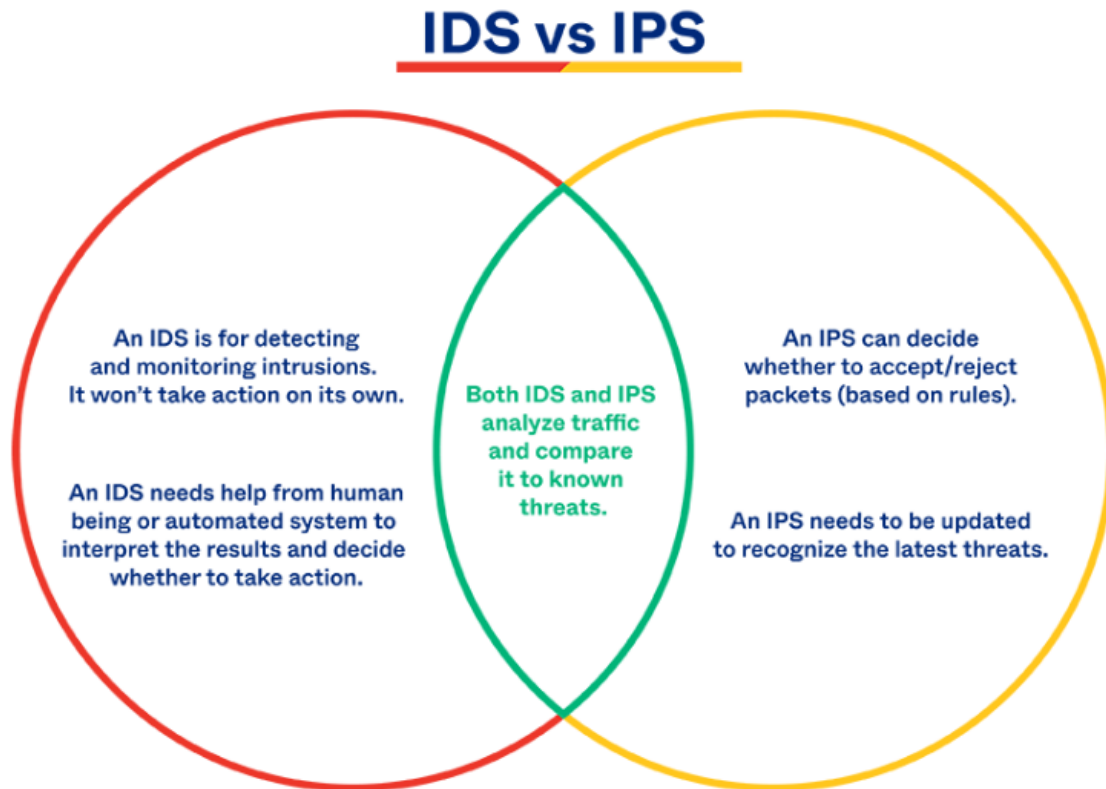


Figure 3. IDS vs IPS

Scan through these lists, and you may immediately understand which system is best for you. If your company can't handle a disruption due to a technical error, for example, an IDS may be best. But if you can't endure even a second of an attack without losing trade secrets or your reputation, an IPS may be better.

Therefore, if you want to protect the assets on your server. But you don't want to slow down the traffic, even if a problem occurs. An intrusion detection system (IDS) could be the solution you've been looking for. But if you want to stop an attack as soon as it's discovered, even if that means closing down legitimate traffic for security concerns. An intrusion protection system (IPS) could be just right for you. let's enumerate the difference in IDS vs IPS Table 1.

Table 1. IDS vs IPS

| PARAMETER                  | IPS                                                                                                                             | IDS                                                                       |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| System Type                | Active (monitor & automatically defend) and/ or passive                                                                         | Passive (monitor and Notify)                                              |
| Detection mechanism        | Statistical anomaly based detection<br>Signature detection:<br>* Exploit-facing signatures<br>* Vulnerability-facing signatures | Signature detection:<br>* Exploit-facing signatures                       |
| Placement                  | Inline to data communication                                                                                                    | Out of band from data communication                                       |
| Anomaly response           | Drop, alert or clean malicious traffic                                                                                          | Sends alarm/alert of detecting malicious traffic                          |
| Network performance impact | Slow down network performance due to delay caused by inline IPS processing                                                      | Does not impact network performance due to non-line deployment of IDS     |
| Benefits                   | Preferred by most organization since detection and prevention are automatically performed                                       | Does not block legitimate traffic which might be blocked by IPS at times. |

## 1.4 IDS & IPS Working Together

Many companies avoid the IDS vs. IPS problem by deploying both solutions to protect their assets and servers. Choose an option like this, and you could use the IPS for active network security while the IDS gives you a deep understanding of how the traffic moves across your network. A solution like this gives you information you can build on to keep your entire system safe. But you'll also avoid letting an attack go unchecked.

## 1.5 How Does IDS/IPS Detect Threats?

IDS/IPS systems detect suspicious or unauthorized activity such as phishing attacks, virus infection and distribution, malware and ransomware installation and download, denial of service (DOS), man-in-the-middle attacks, zero-day attacks, SQL injection, and more. Because of the growth in cloud WAN and mobility, stopping cyber-attacks have become more difficult all while attackers have become more sophisticated in their tactics.

## 1.6 Which is better, IDS or IPS?

An intrusion detection system seeks out anomalous behavior and notifies the network administrator when suspicious activity is detected. An intrusion prevention system automatically triggers remediation workflows to block suspicious activity. The decision over which is better depends on personal preference. Do you want to be given the choice to decide whether to take action or do you want that decision to be made for you?

## 1.7 Can an IPS prevent a DDoS attack?

IPS services aren't suited to defense against DDoS attacks. This is because a DDoS strategy never makes it onto the network where IPSs operate. A DDoS attack sends a flood of malformed connection requests without any intention of ever making a connection. Edge services are a more appropriate mechanism for absorbing DDoS traffic.

## 1.8 Open Source IDS and IPS Tools

### **Snort:**

Snort acts as both an IDS and IPS, has a high level of customization that makes it the choice for a lot of different organizations. This Linux utility is easy to deploy and can be configured to monitor your network traffic for intrusion attempts, log them, and take a specified action when an intrusion attempt is detected. It has the ability to perform real-time traffic analysis and packet logging on Internet Protocol (IP) networks. It uses a series of rules that help define malicious network activity and uses those rules to find packets that match against them and generates alerts for users.

It performs protocol analysis, content searching and matching. The program can also be used to detect probes or attacks, including, but not limited to, operating system fingerprinting attempts, semantic URL attacks, buffer overflows, server message block probes, and stealth port scans. Snort can be configured in three main modes: 1. sniffer, 2. packet logger, and 3. network intrusion detection.

There's no GUI here and it lacks an administrative console, although lots of other open source tools have been created to help out, such as Snorby, BASE and Sguil. These tools provide a web front end to query and analyze alerts coming from Snort IDS.

Major commit version changes:

- 2021 Snort 3.0 stable release (beta 2018)
- 2003 Snort 2.0
- 1998 Snort 1.0

**SolarWinds Security Event Manager:**

Analyzes logs from Windows, Unix, Linux, and Mac OS systems. It manages data collected by Snort, including real-time data. SEM is also an intrusion prevention system, shipping with over 700 rules to shut down malicious activity. An essential tool for improving security, responding to events and achieving compliance. The SolarWinds Security Event Manager controls access to log files, as the name suggests. However, the tool also has network monitoring capabilities. The software package doesn't include a network monitoring facility, but you can add this capability by using the free tool, Snort for network data gathering. This setup gives you two perspectives on intrusion. There are two categories of detection strategies used by IDSs: network-based and host-based.

The instructions to detect signs of intrusion are included with the SolarWinds software package – these are called event correlation rules. You can choose to leave the system to just detect intrusion and block threats manually. You can also activate the IPS functions of the SolarWinds Security Event Manager to get threat remediation performed automatically. CrowdStrike Falcon: A cloud-based endpoint protection platform that includes threat hunting.

**CrowdStrike:**

is an endpoint detection and response system with added interaction with third-party security tools? The system uses security orchestration, automation, and response (SOAR) to improve both threat hunting and threat mitigation.

**Key Features:**

- Hybrid system
- Coordinates on-premises security tools
- Orchestrates threat responses

CrowdStrike Falcon is a cloud platform of security modules and the XDR builds on a couple of other products on the SaaS system. The first of these is an endpoint protection system called CrowdStrike Falcon Prevent – a next-generation anti-virus. The Prevent tool installs on each endpoint. There are versions of this system for Windows, macOS, and Linux. This system is able to continue protecting endpoints even when the network is down.

## **OSSEC**

Excellent host-based intrusion detection system that is free to use. OSSEC is a very popular IPS system. Its detection methodologies are based on examining log files, which makes it a host-based intrusion detection system. The name of this tool stands for 'Open Source HIDS Security' (despite the lack of an 'H' there).

### **Key Features:**

- Free to use
- Highly regarded
- Host-based

The fact that this is an open-source project is great because it also means that the software is free to use. Despite being open-source, OSSEC is actually owned by a company: Trend Micro. The downside of using free software is that you don't get support. The tool is widely used and the OSSEC user community is a great place to get tips and tricks on using the system. However, if you don't want to risk relying on amateur advice for your company software, you can buy a professional support package from Trend Micro.

## **ManageEngine EventLog Analyzer:**

A log file analyzer that searches for evidence of intrusion. ManageEngine is a leading producer of IT network infrastructure monitoring and management solutions. EventLog Analyzer is part of the company's security products. This is a HIDS that focuses on managing and analyzing log files generated by standard applications and operating systems. The tool installs on Windows Server or Linux. It gathers data from those operating systems and also from Mac OS, IBM AIX, HP UX, and Solaris systems. The logs from Windows systems include sources from Windows Server Windows Vista and above and the Windows DHCP Server.

### **Key Features:**

- Manages and analyzes log files
- Auditing for data protection standards compliance

# CHAPTER Two



# CHAPTER TWO

## install Ubuntu 20.04 Desktop On VMware Workstation.

We will install Ubuntu 20.04 desktop on the VMware workstation; after the VM installation is complete, we will install the VMware tools and connect the Ubuntu to the bridge interface; that way, it will act like another host in your local area network.

### Prerequisite:

Before you proceed with the installation, you need to have.

- Ubuntu 20.04 version:  
<https://releases.ubuntu.com/20.04/>
- And VMware workstation pro:  
<https://www.vmware.com/products/workstation-pro/workstation-pro-evaluation.html>

### 2.1 Create the Ubuntu 20.04 virtual machine.

Click on File -> new virtual machine.

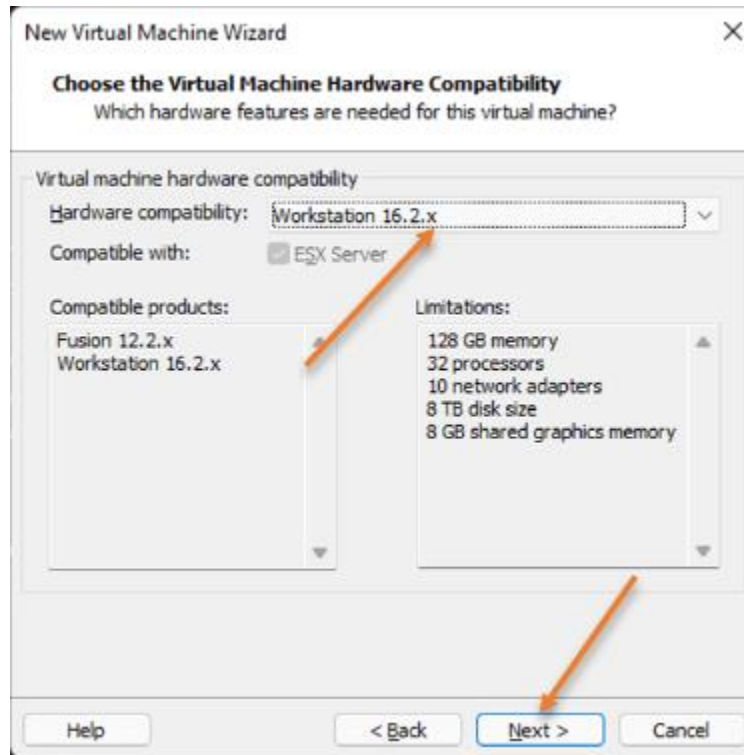
A new virtual machine creation wizard will now open.

Choose a Custom option to proceed.



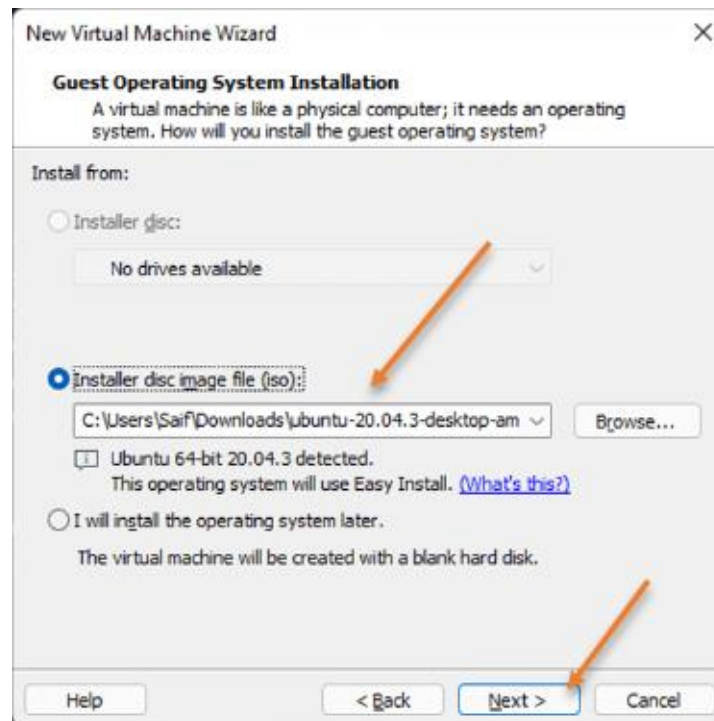
### 2.1.1 Choose the hardware compatibility.

I am using the VMware workstation 16 here, so it does make sense to keep the same version in hardware compatibility. Ubuntu 20.04 is compatible with the 16.0 version, so we should be good. Based on your workstation version, you may make a choice.



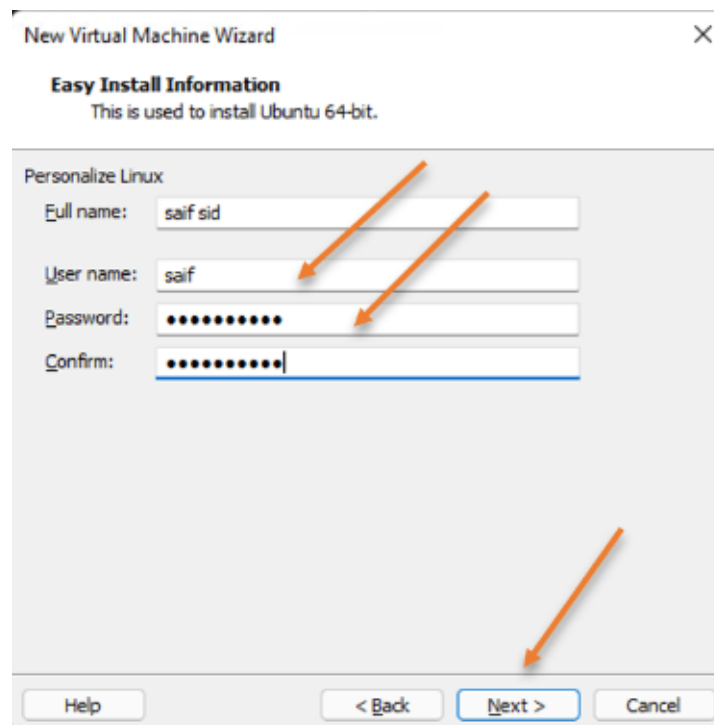
### 2.1.2 Attach the ISO image.

We have already downloaded the Ubuntu 20.04 desktop on our machine; let us attach them under 'Installer disk image file (iso)'; And click on Next.



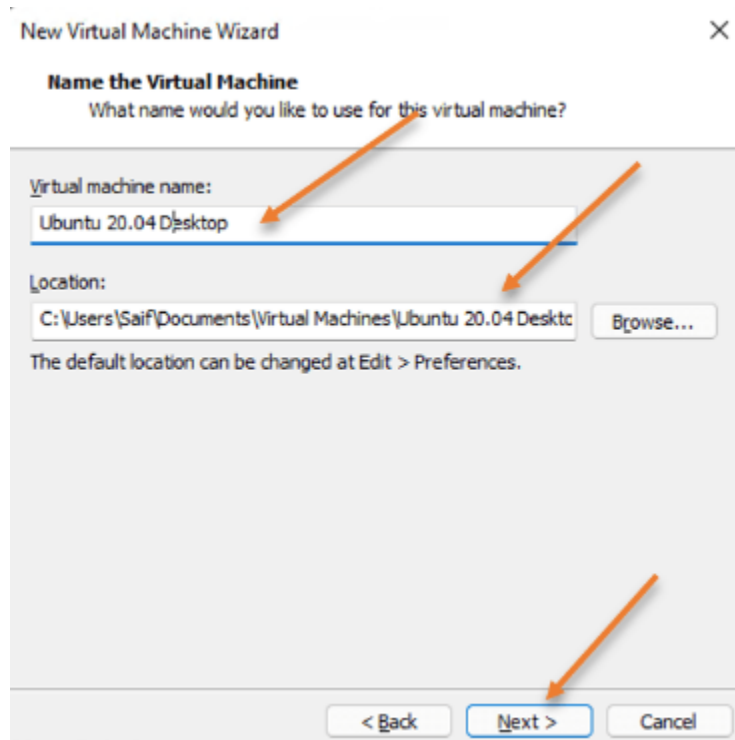
### 2.1 3 Set up the user credential for ubuntu 20.04.

Here you need to set up a username and password for your Ubuntu VM; after the installation is complete, you don't have to set up a credential again, as it will take the credentials we are setting up now. After that, click on Next.



### 2.1.4 Name the VM and the location.

We need to name the virtual machine and the location to keep the virtual machine files. I will name the virtual machine as Ubuntu 20.04 desktop and leave the default storage location.



### 2.1.5 Configure the CPU.

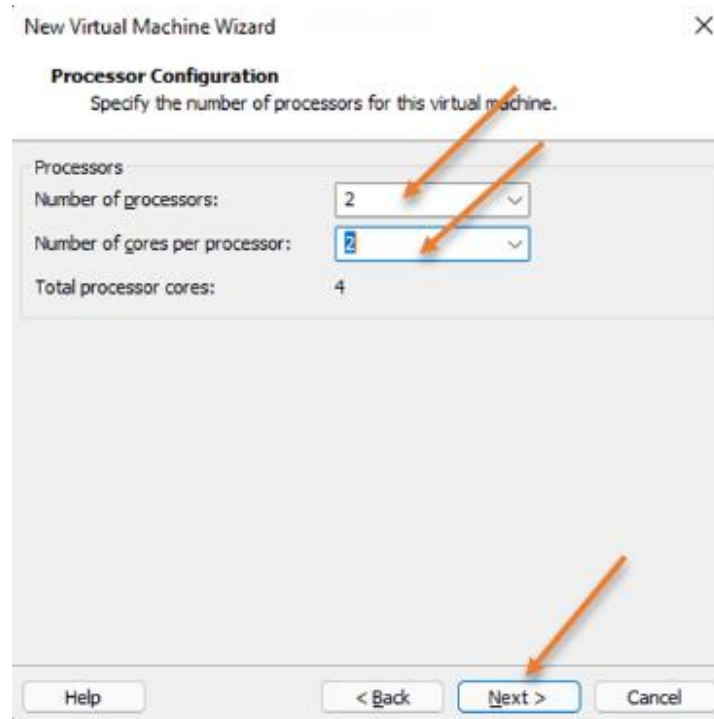
I have about 12 cores of CPU in my PC, so I can safely provide 4 CPUs to the virtual machine as I require good performance.

So choose the Processors as 2.

Select the number of cores per processor as 2, and you will get a total of 4 processors.

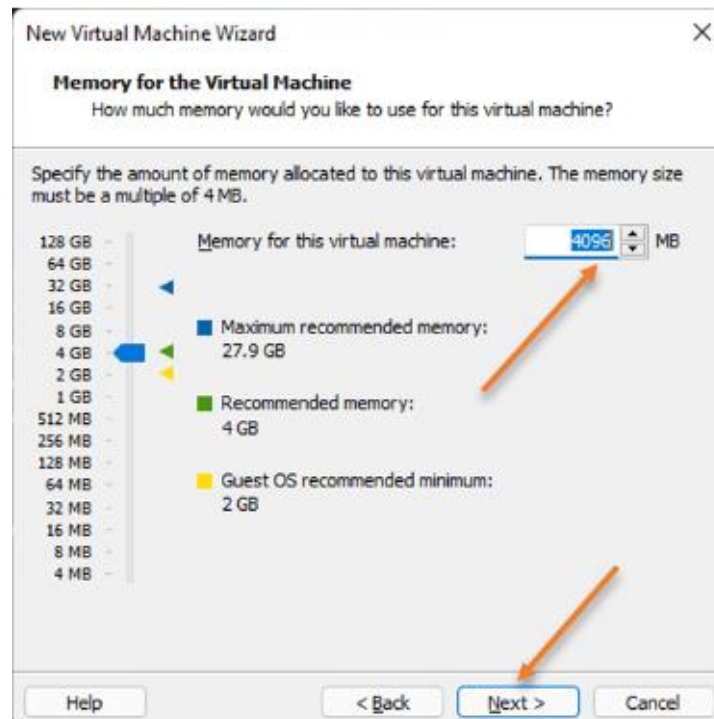
Note: If your machine doesn't have many resources, you can give less than 4 CPUs, as it will be based on the requirement.

After the selection, Click on Next.



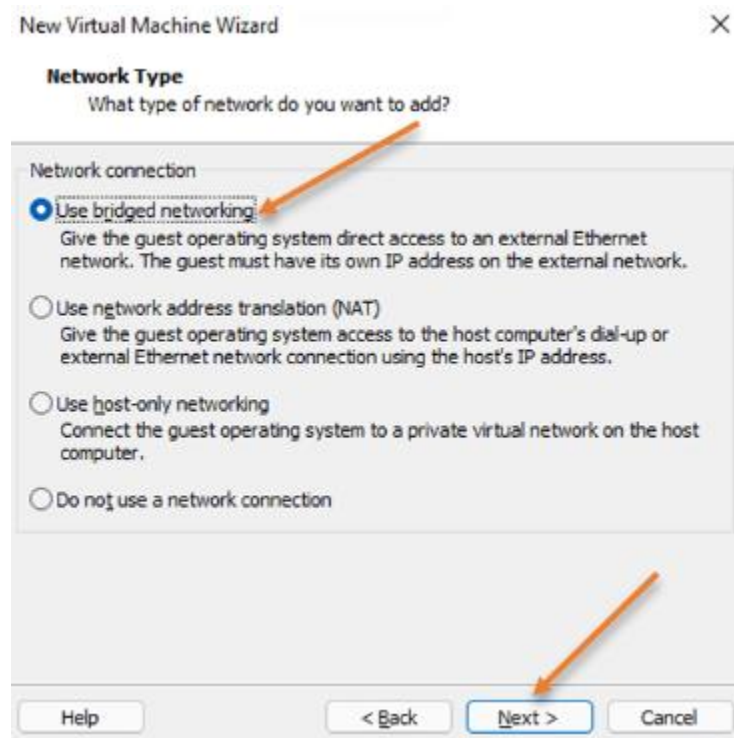
### 2.1.6 Configure the Memory.

If you will use this virtual machine for daily tasks and require you to open multiple applications, I would recommend giving 4GB of RAM. However, 2GB of ram will work just fine as well. Click on next.



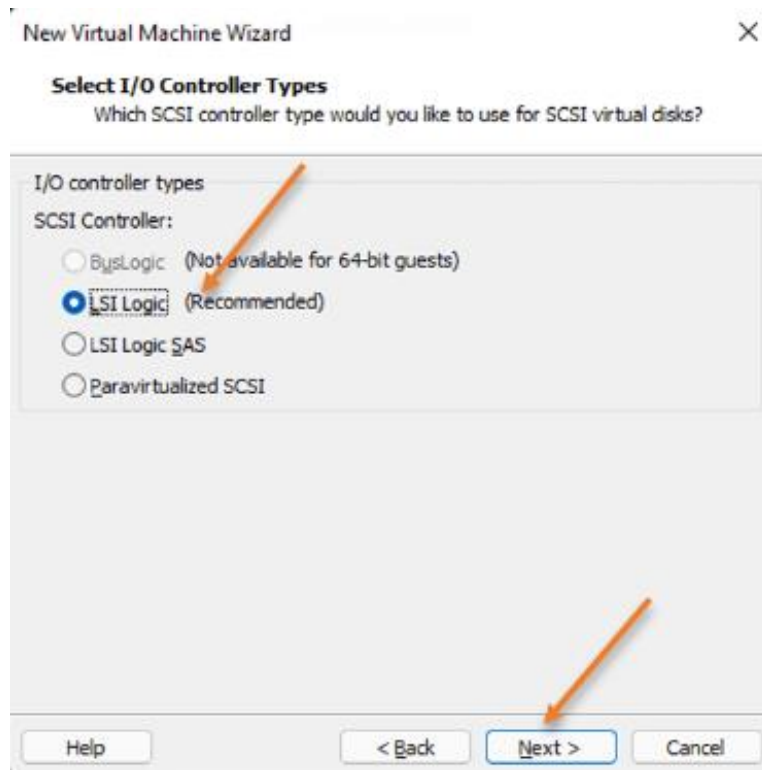
### 2.1.7 Configure the networking.

As I mentioned earlier, I would like this VM to be part of my Local area network, and my router should hand out the IP address to the VM directly. Hence Choose – Use Bridged networking. Click on Next.

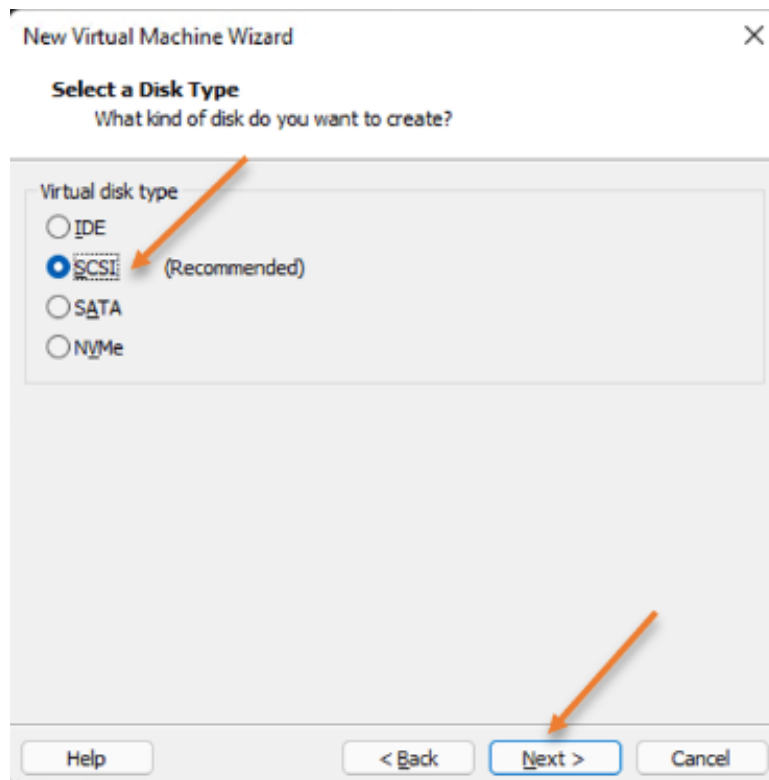


### 2.1.8 Choose the I/O Controller Types.

You can leave the default option here, i.e., the LSI logic, and click Next.

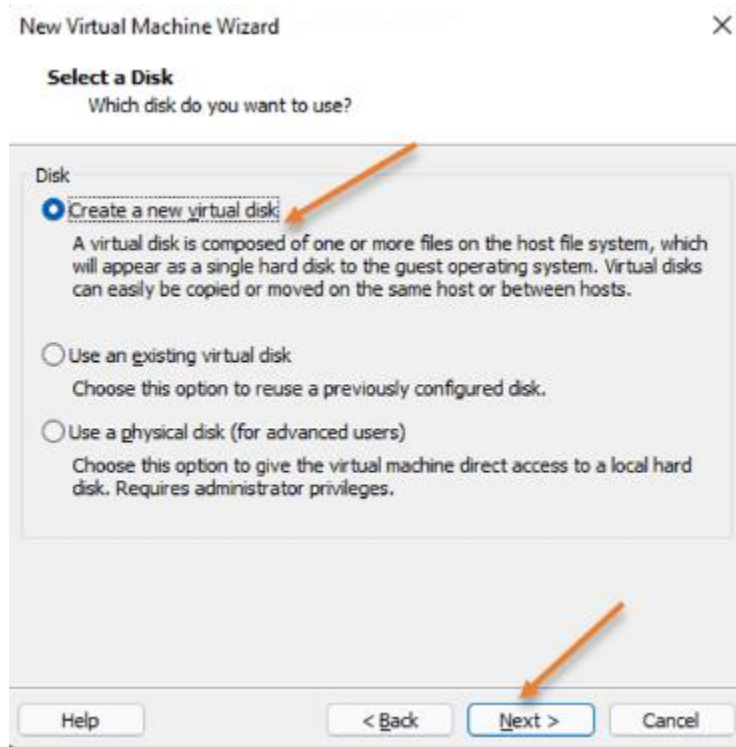


Choose SCSI as the disk type as well.



### 2.1.9 Create a virtual hard disk.

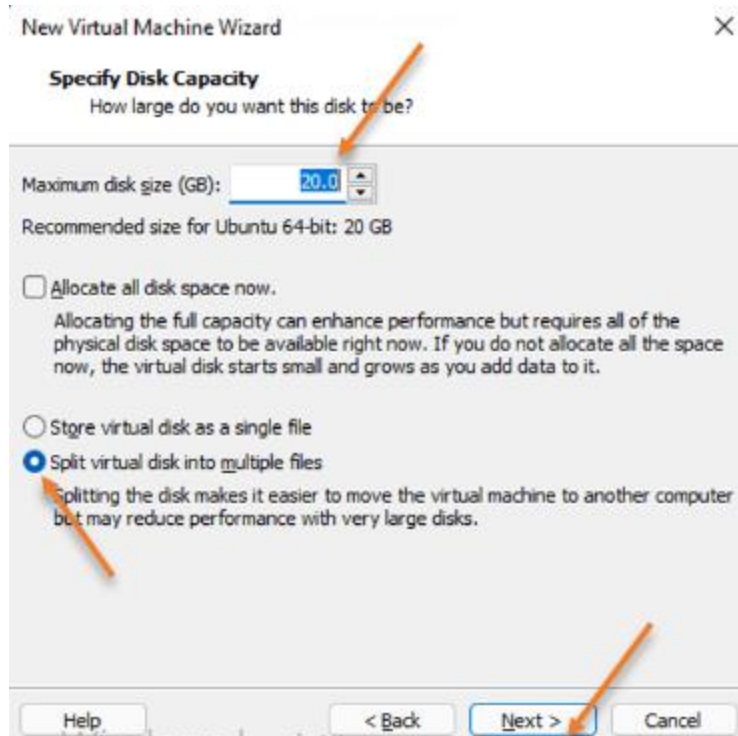
Just like a physical operating system requires a hard Disk, your virtual machine also requires a virtual hard disk to function. Under disk, Choose Create a new virtual hard disk and click Next.



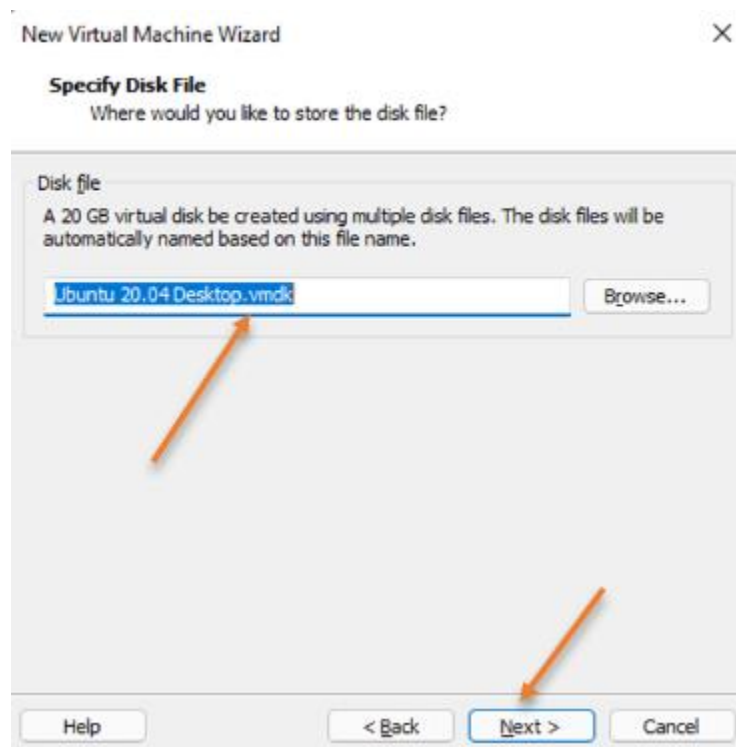
### 2.1.10 Define the storage size.

By default, VMware would assign 20GB of storage to the Ubuntu VM, and you can either increase or decrease the storage depending on your need. I am okay with the 20GB of storage and choose Split virtual disk into multiple files and click Next.

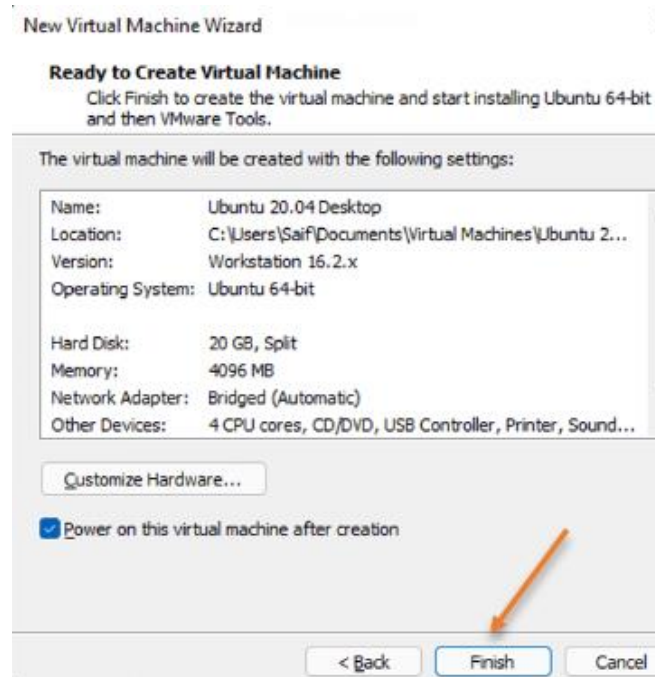




You will be able to change the location of the virtual Hard disk files for your Ubuntu VM; if you want to make changes, you can do that else, leave the default and click Next.



On the ready to create virtual machine window, you will get the properties of the ubuntu virtual machine that we will build, verify whether everything looks perfect, leave the power on the Virtual machine after creation checked, and click on Finish.

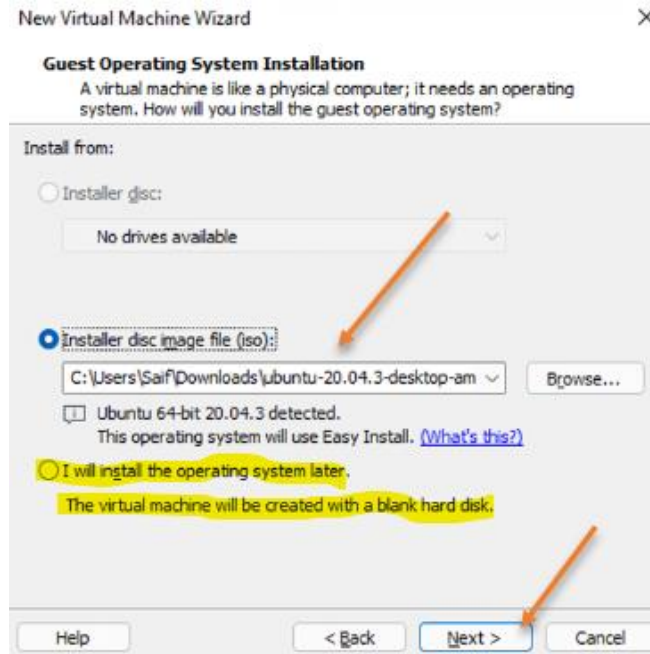


The Ubuntu 20.04 installation will begin; wait for a few seconds for the installation screen to load, after the Ubuntu will proceed with the installation automatically. The installation is now complete, and you will be prompted to enter the credentials.



## 2.2 Installing Empty Copy of Operating System:

We should repeat all stages above. Except, in stage 2 we don't attach the iso file of the operating system, instead we choose the choice: install the operating system later as the figure below:

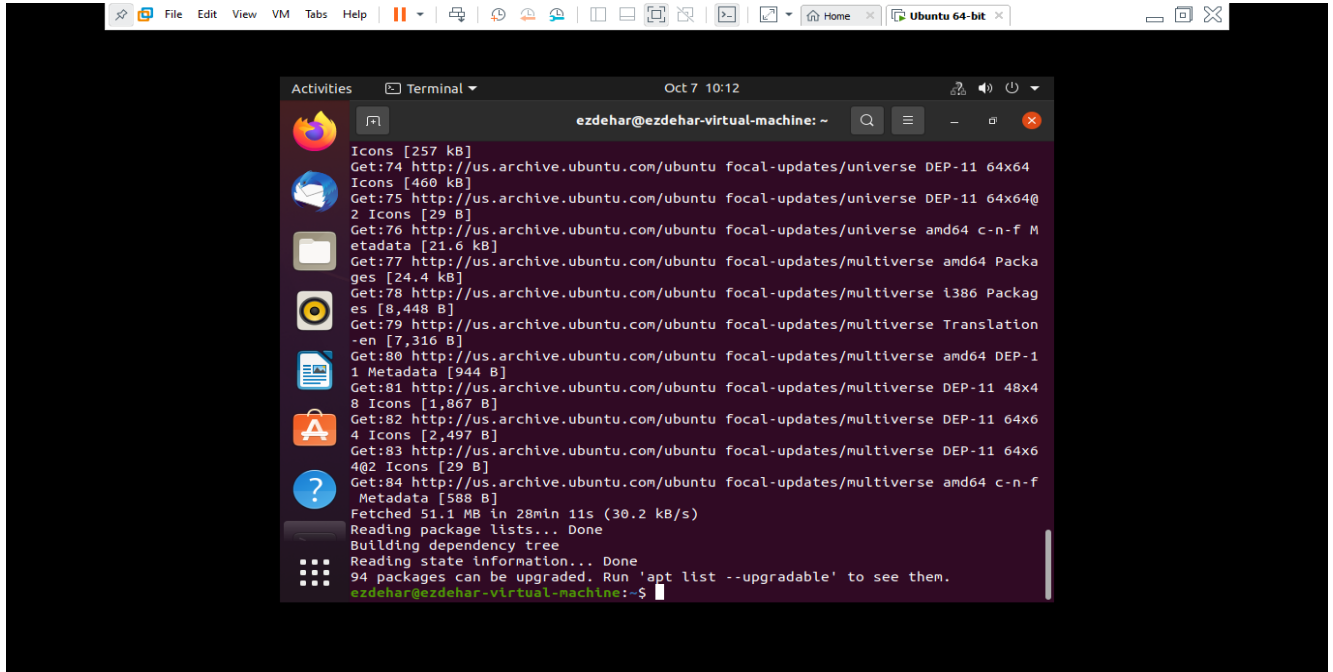


## 2.3 Installing SNORT

2.3.1 Update: **Virtual-machine:~** \$ sudo apt-get update

Or sudo apt update

**OUTPUT:**

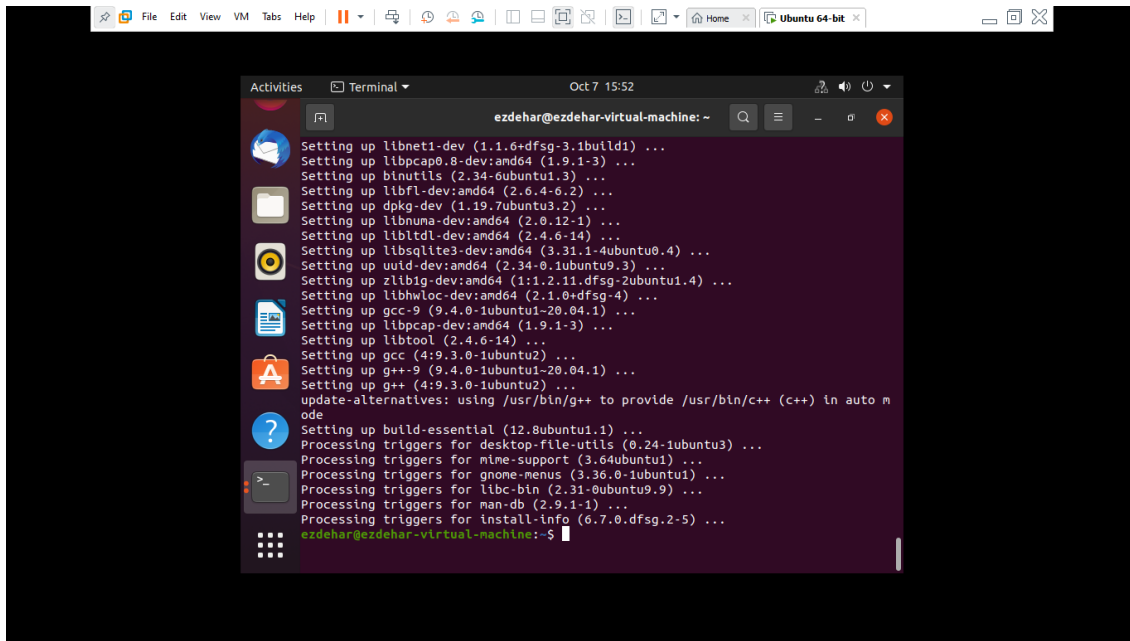
A screenshot of a terminal window within a virtual machine environment. The terminal title bar shows 'ezdehar@ezdehar-virtual-machine: ~'. The output of the 'sudo apt-get update' command is displayed, showing progress for various Ubuntu repositories. It lists the size of icons, the number of packages to be fetched, and the total size of the fetched data. The output concludes with '94 packages can be upgraded. Run 'apt list --upgradable' to see them.'

```
Icons [257 kB]
Get:74 http://us.archive.ubuntu.com/ubuntu focal-updates/universe DEP-11 64x64
Icons [460 kB]
Get:75 http://us.archive.ubuntu.com/ubuntu focal-updates/universe DEP-11 64x64@
2 Icons [29 B]
Get:76 http://us.archive.ubuntu.com/ubuntu focal-updates/universe amd64 c-n-f M
etadata [21.6 kB]
Get:77 http://us.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 Packa
ges [24.4 kB]
Get:78 http://us.archive.ubuntu.com/ubuntu focal-updates/multiverse i386 Packag
es [8,448 B]
Get:79 http://us.archive.ubuntu.com/ubuntu focal-updates/multiverse Translation
-en [7,316 B]
Get:80 http://us.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 DEP-1
1 Metadata [944 B]
Get:81 http://us.archive.ubuntu.com/ubuntu focal-updates/multiverse DEP-11 48x4
8 Icons [1,867 B]
Get:82 http://us.archive.ubuntu.com/ubuntu focal-updates/multiverse DEP-11 64x6
4 Icons [2,497 B]
Get:83 http://us.archive.ubuntu.com/ubuntu focal-updates/multiverse DEP-11 64x6
4@2 Icons [29 B]
Get:84 http://us.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 c-n-f
Metadata [588 B]
Fetched 51.1 MB in 28min 11s (30.2 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
94 packages can be upgraded. Run 'apt list --upgradable' to see them.
ezdehar@ezdehar-virtual-machine:~$
```

2.3.2 install build tools for snort:

**Virtual-machine:~** \$ sudo apt-get install build-essential libpcap-dev  
libpcrc3-dev libnet1-dev zlib1g-dev luajit hwloc libdnet-dev libdumbnet-dev  
bison flex liblzma-dev openssl libssl-dev pkg-config libhwloc-dev cmake  
cpputest libsqlite3-dev uuid-dev libcmocka-dev libnetfilter-queue-dev  
libmnl-dev autotools-dev liblua5.1-dev libunwind-dev

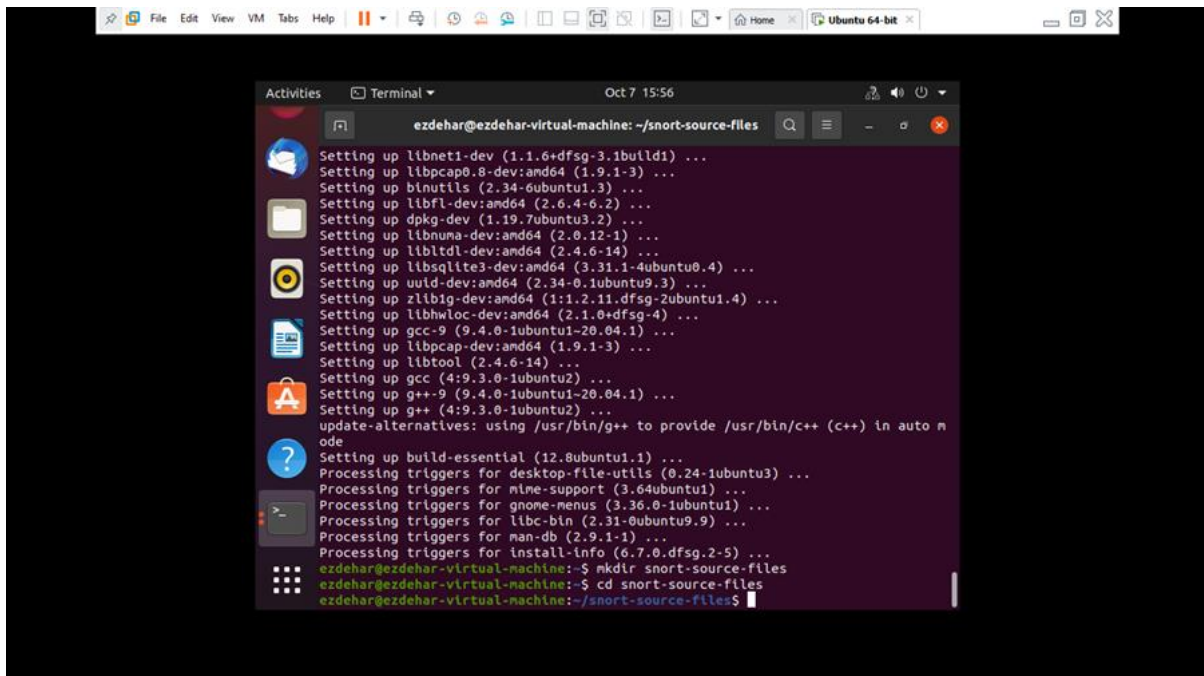
**OUTPUT:**



2.3.3 create directory where kept source files for snort:

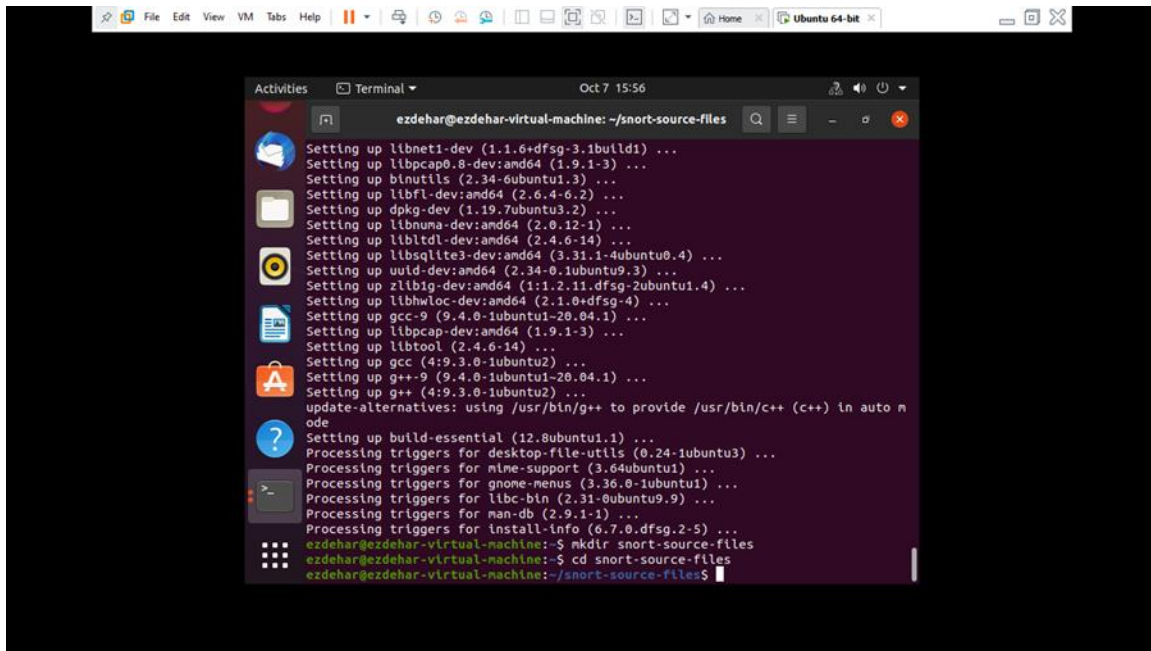
**Virtual-machine:~ \$** mkdir snort-source-files

**OUTPUT:**



### 2.3.4 Open directory: **Virtual-machine:~\$ cd snort-source-files**

## OUTPUT:

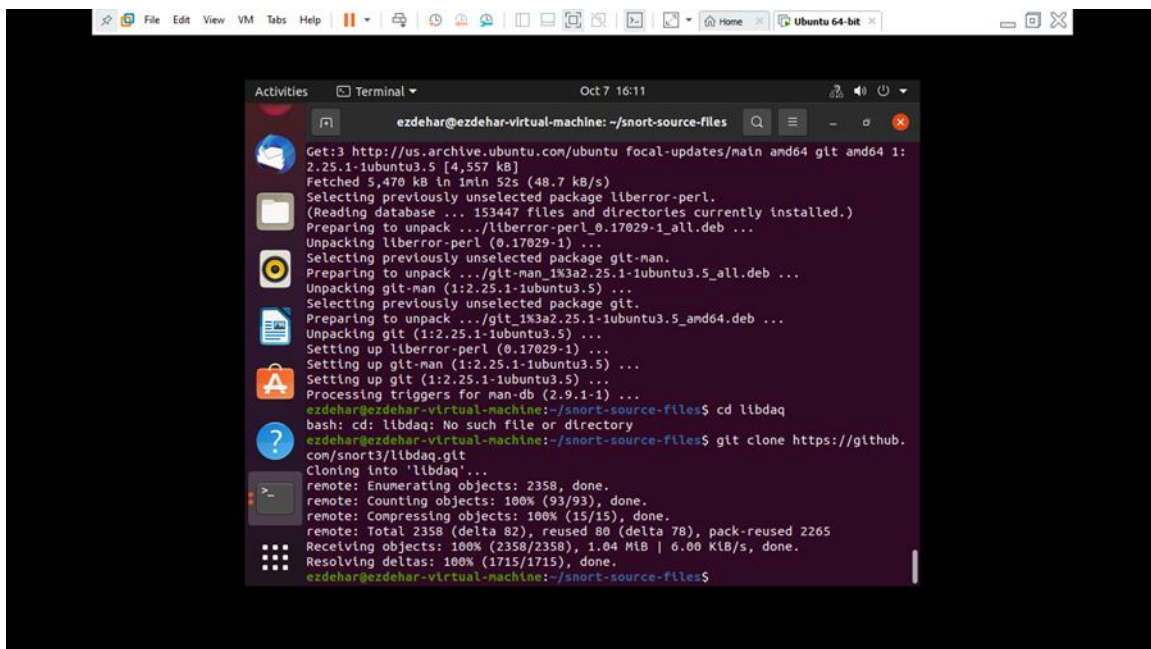


```
Setting up libnet1-dev (1.1.6+dfsg-3.1build1) ...
Setting up libpcap0.8-dev:amd64 (1.9.1-3) ...
Setting up binutils (2.34-6ubuntu1.3) ...
Setting up libfl-dev:amd64 (2.6.4-6.2) ...
Setting up dpkg-dev (1.19.7ubuntu3.2) ...
Setting up libnmap-dev:amd64 (2.0.12-1) ...
Setting up libltdl-dev:amd64 (2.4.6-14) ...
Setting up libsqlite3-dev:amd64 (3.31.1-4ubuntu0.4) ...
Setting up uuid-dev:amd64 (2.34-0.1ubuntu9.3) ...
Setting up zlib1g-dev:amd64 (1:1.2.11.dfsg-2ubuntu1.4) ...
Setting up libhwloc-dev:amd64 (2.1.0+dfsg-4) ...
Setting up gcc-9 (9.4.0-1ubuntu1-20.04.1) ...
Setting up libpcap-dev:amd64 (1.9.1-3) ...
Setting up libtool (2.4.6-14) ...
Setting up gcc (4:9.3.0-1ubuntu2) ...
Setting up g++-9 (9.4.0-1ubuntu1-20.04.1) ...
Setting up g++ (4:9.3.0-1ubuntu2) ...
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mode
Setting up build-essential (12.8ubuntu1.1) ...
Processing triggers for desktop-file-utils (0.24-1ubuntu3) ...
Processing triggers for mime-support (3.64ubuntu1) ...
Processing triggers for gnome-menus (3.36.0-1ubuntu1) ...
Processing triggers for libc-bin (2.31-0ubuntu9.9) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for install-info (6.7.0.dfsg.2-5) ...
ezdehar@ezdehar-virtual-machine: ~/snort-source-files$ cd snort-source-files
ezdehar@ezdehar-virtual-machine: ~/snort-source-files$
```

### 2.3.5 Install lipDAQ: **Virtual-machine:~/snort-source-files\$ git clone**

<https://github.com/snort3/libdaq.git>

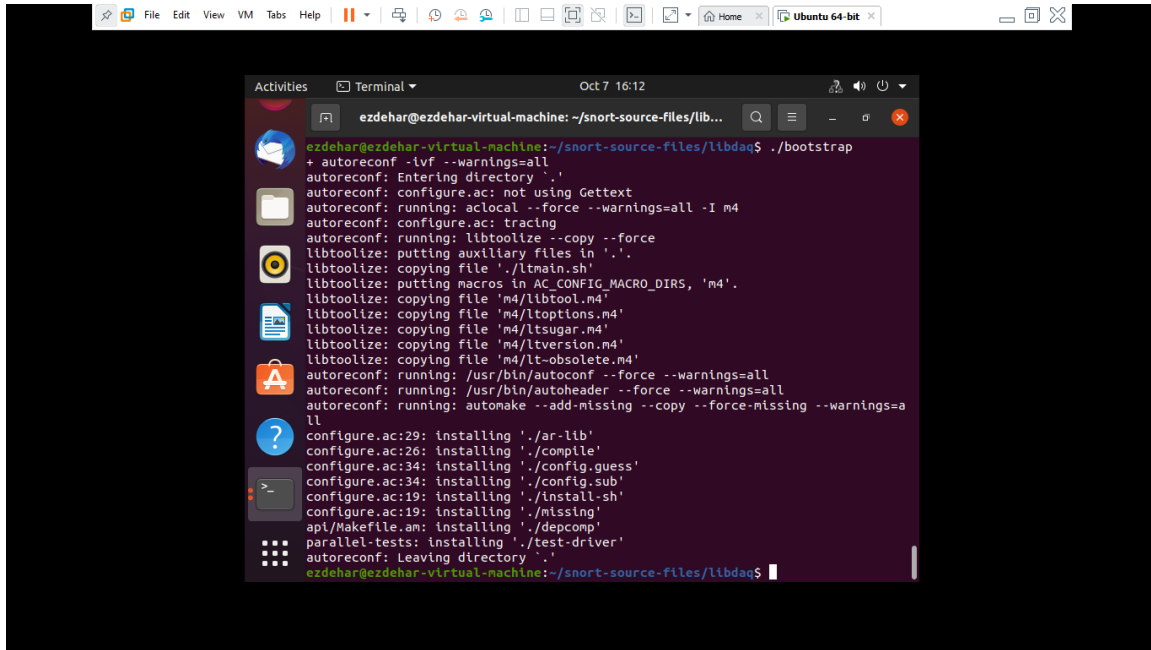
## OUTPUT:



```
Get:3 http://us.archive.ubuntu.com/ubuntu focal-updates/main amd64 git amd64 1:
2.25.1-1ubuntu3.5 [4,557 kB]
Fetched 5,470 kB in 1min 52s (48.7 kB/s)
Selecting previously unselected package liberror-perl.
(Reading database ... 153447 files and directories currently installed.)
Preparing to unpack .../liberror-perl_0.17029-1_all.deb ...
Unpacking liberror-perl (0.17029-1) ...
Selecting previously unselected package git-man.
Preparing to unpack .../git-man_1%3a2.25.1-1ubuntu3.5_all.deb ...
Unpacking git-man (1:2.25.1-1ubuntu3.5) ...
Selecting previously unselected package git.
Preparing to unpack .../git_1%3a2.25.1-1ubuntu3.5_amd64.deb ...
Unpacking git (1:2.25.1-1ubuntu3.5) ...
Setting up liberror-perl (0.17029-1) ...
Setting up git-man (1:2.25.1-1ubuntu3.5) ...
Setting up git (1:2.25.1-1ubuntu3.5) ...
Processing triggers for man-db (2.9.1-1) ...
ezdehar@ezdehar-virtual-machine: ~/snort-source-files$ cd libdaq
bash: cd: libdaq: No such file or directory
ezdehar@ezdehar-virtual-machine: ~/snort-source-files$ git clone https://github.
com/snort3/libdaq.git
Cloning into 'libdaq'...
remote: Enumerating objects: 2358, done.
remote: Counting objects: 100% (93/93), done.
remote: Compressing objects: 100% (15/15), done.
remote: Total 2358 (delta 82), reused 80 (delta 78), pack-reused 2265
Receiving objects: 100% (2358/2358), 1.04 MiB | 6.00 KiB/s, done.
Resolving deltas: 100% (1715/1715), done.
ezdehar@ezdehar-virtual-machine: ~/snort-source-files$
```

2.3.6 Open directory: **Virtual-machine:~/snort-source-files\$ cd libdaq**  
**7-Install bootstrap: Virtual-machine:~/snort-source-files/libdaq\$**  
**./bootstrap**

## OUTPUT:

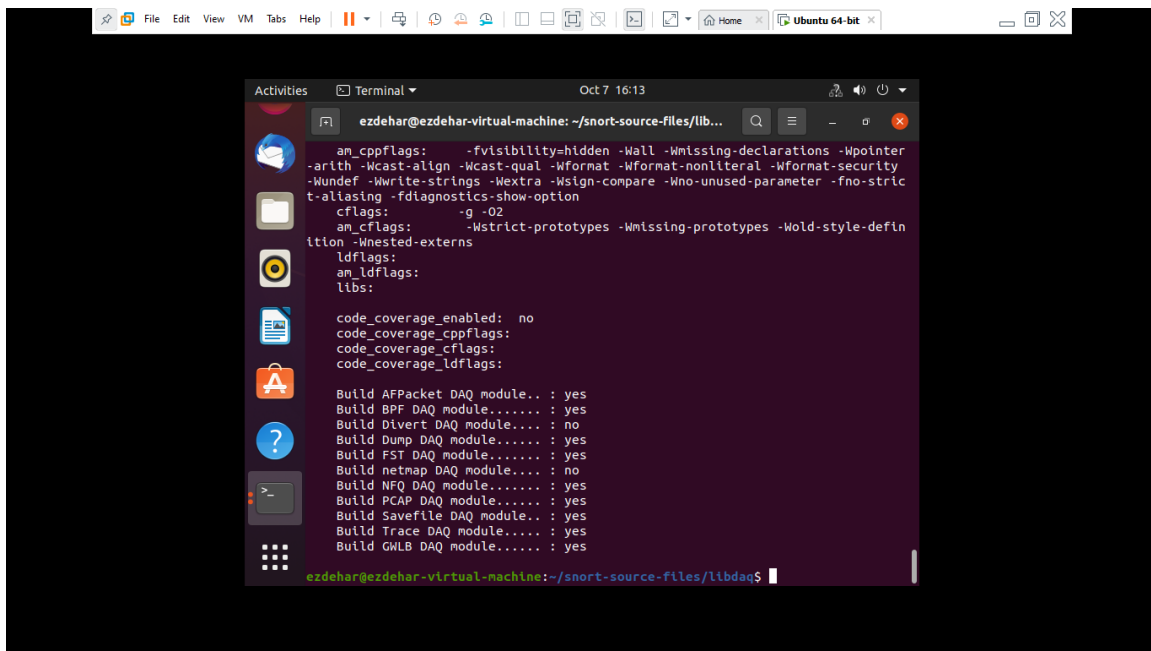


```
ezdehar@ezdehar-virtual-machine: ~/snort-source-files/libdaq$ ./bootstrap
+ autoreconf -lvf --warnings=all
autoreconf: Entering directory '.'
autoreconf: configure.ac: not using Gettext
autoreconf: running: aclocal --force --warnings=all -I m4
autoreconf: configure.ac: tracing
autoreconf: running: libtoolize --copy --force
libtoolize: putting auxiliary files in '.'
libtoolize: copying file './ltmain.sh'
libtoolize: putting macros in AC_CONFIG_MACRO_DIRS, 'm4'.
libtoolize: copying file 'm4/libtool.m4'
libtoolize: copying file 'm4/ltoptions.m4'
libtoolize: copying file 'm4/ltversion.m4'
libtoolize: copying file 'm4/ltobsolete.m4'
autoreconf: running: /usr/bin/autoconf --force --warnings=all
autoreconf: running: /usr/bin/autoheader --force --warnings=all
autoreconf: running: automake --add-missing --copy --force-missing --warnings=all
configure.ac:29: installing './ar-lib'
configure.ac:26: installing './compile'
configure.ac:34: installing './config.guess'
configure.ac:34: installing './config.sub'
configure.ac:19: installing './install-sh'
configure.ac:19: installing './missing'
apl/Makefile.am: installing './depcomp'
parallel-tests: installing './test-driver'
autoreconf: Leaving directory '.'
ezdehar@ezdehar-virtual-machine:~/snort-source-files/libdaq$
```

2.3.7 Install configuration: **Virtual-machine:~/snort-source-files/libdaq\$**  
**./configure**

## OUTPUT:





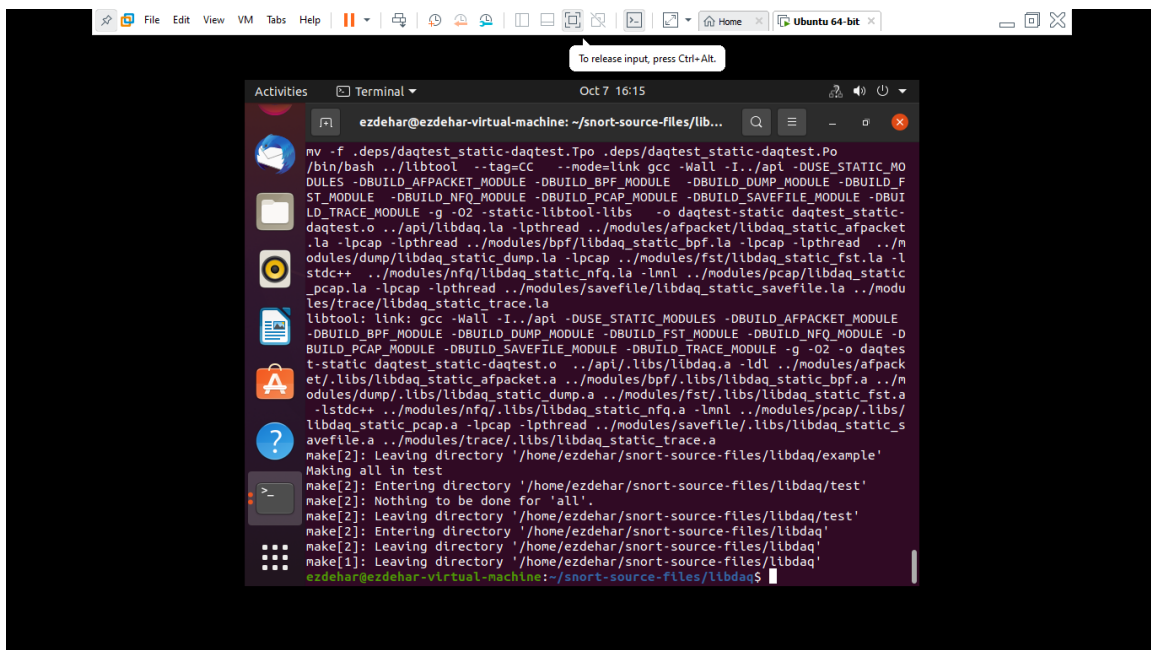
```
am_cppflags: -fvvisibility=hidden -Wall -Wmissing-declarations -Wpointer
-arith -Wcast-align -Wcast-qual -Wformat -Wformat-nonliteral -Wformat-security
-Wundef -Wwrite-strings -Wextra -Wsign-compare -Wno-unused-parameter -fno-strict-aliasing -fdiagnostics-show-option
cflags: -g -O2
am_cflags: -Wstrict-prototypes -Wmissing-prototypes -Wold-style-definition -Wnested-externs
ldflags:
am_ldflags:
libs:

code_coverage_enabled: no
code_coverage_cppflags:
code_coverage_cflags:
code_coverage_ldflags:

Build AFPPacket DAQ module... : yes
Build BPF DAQ module..... : yes
Build Divert DAQ module.... : no
Build Dump DAQ module..... : yes
Build FST DAQ module..... : yes
Build netmap DAQ module.... : no
Build NFQ DAQ module..... : yes
Build PCAP DAQ module..... : yes
Build Savefile DAQ module.. : yes
Build Trace DAQ module.... : yes
Build GMLB DAQ module..... : yes

ezdehar@ezdehar-virtual-machine:~/snort-source-files/libdaq$
```

2.3.8 Make: **Virtual-machine:~/snort-source-files/libdaq\$ make**  
**OUTPUT:**



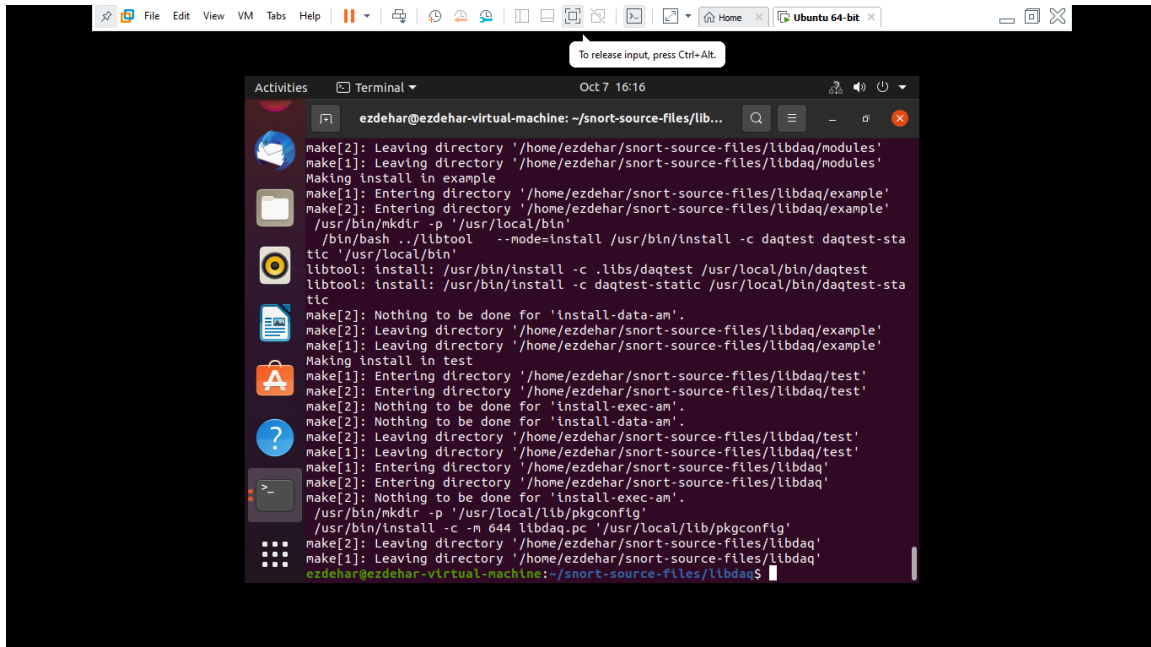
```
mv -f .deps/daqtest_static-daqtest.Tpo .deps/daqtest_static-daqtest.Po
/bin/bash ./libtool --tag=CC --mode=link gcc -Wall -I../api -DUSE_STATIC_MODULES -DBUILD_AFPACKET_MODULE -DBUILD_BPF_MODULE -DBUILD_DUMP_MODULE -DBUILD_FST_MODULE -DBUILD_NFQ_MODULE -DBUILD_PCAP_MODULE -DBUILD_SAVEFILE_MODULE -DBUILD_TRACE_MODULE -g -O2 -static -libtool-libs -o daqtest_static daqtest_static-daqtest.o ../api/libdaq_la -lpthread ../modules/afppacket/libdaq_static_afppacket_la -lpcap -lpthread ../modules/bpf/libdaq_static_bpf_la -lpcap -lpthread ../modules/dump/libdaq_static_dump_la -lpcap ../modules/fst/libdaq_static_fst_la -lstdc++ ../modules/nfq/libdaq_static_nfq_la -lnnl ../modules/pcap/libdaq_static_pcap_la -lpcap -lpthread ../modules/savefile/libdaq_static_savefile_la ../modules/trace/libdaq_static_trace_la
libtool: link: gcc -Wall -I../api -DUSE_STATIC_MODULES -DBUILD_AFPACKET_MODULE -DBUILD_BPF_MODULE -DBUILD_DUMP_MODULE -DBUILD_FST_MODULE -DBUILD_NFQ_MODULE -DBUILD_PCAP_MODULE -DBUILD_SAVEFILE_MODULE -DBUILD_TRACE_MODULE -g -O2 -o daqtest_static daqtest_static-daqtest.o ../api/.libs/libdaq.a -ldl ../modules/afppacket/.libs/libdaq_static_afppacket.a ../modules/bpf/.libs/libdaq_static_bpf.a ../modules/dump/.libs/libdaq_static_dump.a ../modules/fst/.libs/libdaq_static_fst.a -lstdc++ ../modules/nfq/.libs/libdaq_static_nfq.a -lnnl ../modules/pcap/.libs/libdaq_static_pcap.a -lpcap -lpthread ../modules/savefile/.libs/libdaq_static_savefile.a ../modules/trace/.libs/libdaq_static_trace.a
make[2]: Leaving directory '/home/ezdehar/snort-source-files/libdaq/example'
Making all in test
make[2]: Entering directory '/home/ezdehar/snort-source-files/libdaq/test'
make[2]: Nothing to be done for 'all'.
make[2]: Leaving directory '/home/ezdehar/snort-source-files/libdaq/test'
make[2]: Entering directory '/home/ezdehar/snort-source-files/libdaq'
make[2]: Leaving directory '/home/ezdehar/snort-source-files/libdaq'
make[1]: Leaving directory '/home/ezdehar/snort-source-files/libdaq'

ezdehar@ezdehar-virtual-machine:~/snort-source-files/libdaq$
```

2.3.9 Make install: **Virtual-machine:~/snort-source-files/libdaq\$ make install**



## OUTPUT:

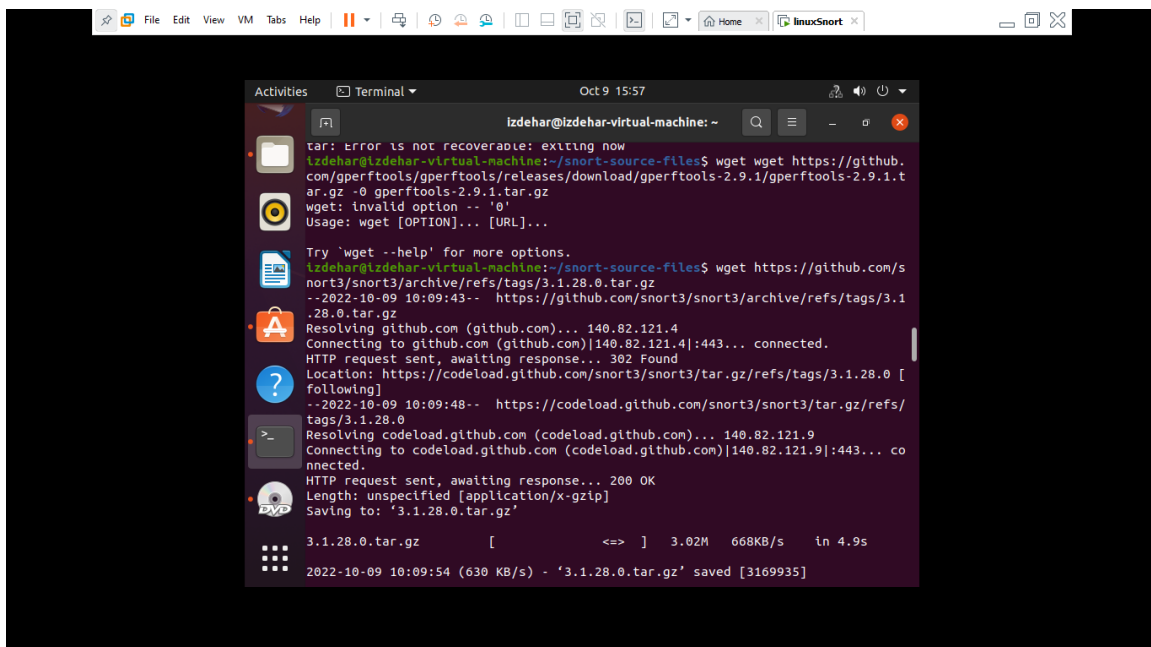


```
ezdehar@ezdehar-virtual-machine: ~/snort-source-files/lib...
make[2]: Leaving directory '/home/ezdehar/snort-source-files/libdaq/modules'
make[1]: Leaving directory '/home/ezdehar/snort-source-files/libdaq/modules'
Making install in example
make[1]: Entering directory '/home/ezdehar/snort-source-files/libdaq/example'
make[2]: Entering directory '/home/ezdehar/snort-source-files/libdaq/example'
/usr/bin/mkdir -p /usr/local/bin
/bin/bash ./libtool --mode=install /usr/bin/install -c daqtest daqtest-static /usr/local/bin
libtool: install: /usr/bin/install -c .libs/daqtest /usr/local/bin/daqtest
libtool: install: /usr/bin/install -c daqtest-static /usr/local/bin/daqtest-static
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/home/ezdehar/snort-source-files/libdaq/example'
make[1]: Leaving directory '/home/ezdehar/snort-source-files/libdaq/example'
Making install in test
make[1]: Entering directory '/home/ezdehar/snort-source-files/libdaq/test'
make[2]: Entering directory '/home/ezdehar/snort-source-files/libdaq/test'
make[2]: Nothing to be done for 'install-exec-am'.
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/home/ezdehar/snort-source-files/libdaq/test'
make[1]: Leaving directory '/home/ezdehar/snort-source-files/libdaq/test'
make[1]: Entering directory '/home/ezdehar/snort-source-files/libdaq'
make[2]: Entering directory '/home/ezdehar/snort-source-files/libdaq'
make[2]: Nothing to be done for 'install-exec-am'.
/usr/bin/mkdir -p /usr/local/lib/pkgconfig
/usr/bin/install -c -m 644 libdaq.pc /usr/local/lib/pkgconfig
make[2]: Leaving directory '/home/ezdehar/snort-source-files/libdaq'
make[1]: Leaving directory '/home/ezdehar/snort-source-files/libdaq'
ezdehar@ezdehar-virtual-machine:~/snort-source-files/libdaq$
```

2.3.10 return to source directory: **Virtual-machine:~/snort-source-files/libdaq\$** `cd ../`

2.3.11 install snort3: **Virtual-machine:~/snort-source-files\$** `wget https://github.com/snort3/snort3/archive/refs/tags/3.1.28.0.tar.gz`

## OUTPUT:



```
lzdehar@lzdehar-virtual-machine: ~
tar: Error is not recoverable: exiting now
lzdehar@lzdehar-virtual-machine:~/snort-source-files$ wget wget https://github.com/gperftools/gperftools/releases/download/gperftools-2.9.1/gperftools-2.9.1.tar.gz
ar.gz -O gperftools-2.9.1.tar.gz
wget: invalid option -- '0'
Usage: wget [OPTION]... [URL]...

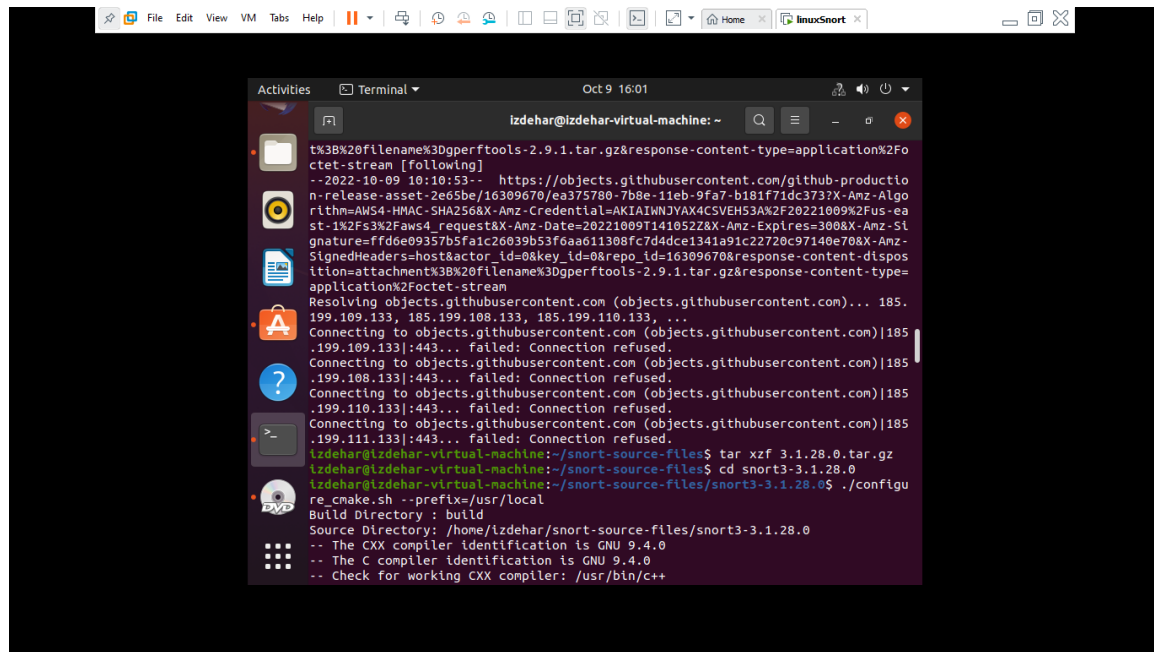
Try 'wget --help' for more options.
lzdehar@lzdehar-virtual-machine:~/snort-source-files$ wget https://github.com/snort3/snort3/archive/refs/tags/3.1.28.0.tar.gz
--2022-10-09 10:09:43-- https://github.com/snort3/snort3/archive/refs/tags/3.1.28.0.tar.gz
Resolving github.com (github.com)... 140.82.121.4
Connecting to github.com (github.com)|140.82.121.4|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://codeload.github.com/snort3/snort3/tar.gz/refs/tags/3.1.28.0 [following]
--2022-10-09 10:09:48-- https://codeload.github.com/snort3/snort3/tar.gz/refs/tags/3.1.28.0
Resolving codeload.github.com (codeload.github.com)... 140.82.121.9
Connecting to codeload.github.com (codeload.github.com)|140.82.121.9|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [application/x-gzip]
Saving to: '3.1.28.0.tar.gz'

3.1.28.0.tar.gz      [          ]  3.02M  668KB/s  in 4.9s
2022-10-09 10:09:54 (630 KB/s) - '3.1.28.0.tar.gz' saved [3169935]
```

2.3.12 Open files snort3: **Virtual-machine:~/snort-source-files\$ tar xzf 3.1.28.0.tar.gz**

2.3.13 Open directory: **Virtual-machine:~/snort-source-files\$ cd snort3-3.1.28.0**

**OUTPUT:**



```
izdehar@izdehar-virtual-machine: ~  
t%3B%20filename%3Dgperftools-2.9.1.tar.gz&response-content-type=application%2Foctet-stream [following]  
--2022-10-09 10:10:53-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/16309670/ea375780-7b8e-11eb-9fa7-b181f71dc373?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=AKIAIWNJYAX4CSVEH53A%2F20221009%2Fus-east-1%2Fus3%2Faws4_request&X-Amz-Date=20221009T141052Z&X-Amz-Expires=300&X-Amz-Signature=fdd6e9357b5fa1c26039b53f6a611308fc7d4dce1341e91c22720e97140e708&X-Amz-SignedHeaders=host&actor_id=0&key_id=0&repo_id=16309670&response-content-disposition=attachment%3B%20filename%3Dgperftools-2.9.1.tar.gz&response-content-type=application%2Foctet-stream  
Resolving objects.githubusercontent.com (objects.githubusercontent.com)... 185.199.109.133, 185.199.108.133, 185.199.110.133, ...  
Connecting to objects.githubusercontent.com (objects.githubusercontent.com)|185.199.109.133|:443... failed: Connection refused.  
Connecting to objects.githubusercontent.com (objects.githubusercontent.com)|185.199.108.133|:443... failed: Connection refused.  
Connecting to objects.githubusercontent.com (objects.githubusercontent.com)|185.199.110.133|:443... failed: Connection refused.  
Connecting to objects.githubusercontent.com (objects.githubusercontent.com)|185.199.111.133|:443... failed: Connection refused.  
izdehar@izdehar-virtual-machine:~/snort-source-files$ tar xzf 3.1.28.0.tar.gz  
izdehar@izdehar-virtual-machine:~/snort-source-files$ cd snort3-3.1.28.0  
izdehar@izdehar-virtual-machine:~/snort-source-files/snort3-3.1.28.0$ ./configure_cmake.sh --prefix=/usr/local  
Build Directory : build  
Source Directory: /home/izdehar/snort-source-files/snort3-3.1.28.0  
-- The CXX compiler identification is GNU 9.4.0  
-- The C compiler identification is GNU 9.4.0  
-- Check for working CXX compiler: /usr/bin/c++
```

2.3.14 install configure and make direcotry: **Virtual-machine:~/snort-source-files/ snort3-3.1.28.0\$ ./configure\_cmake.sh --prefix=/usr/local**

**OUTPUT:**

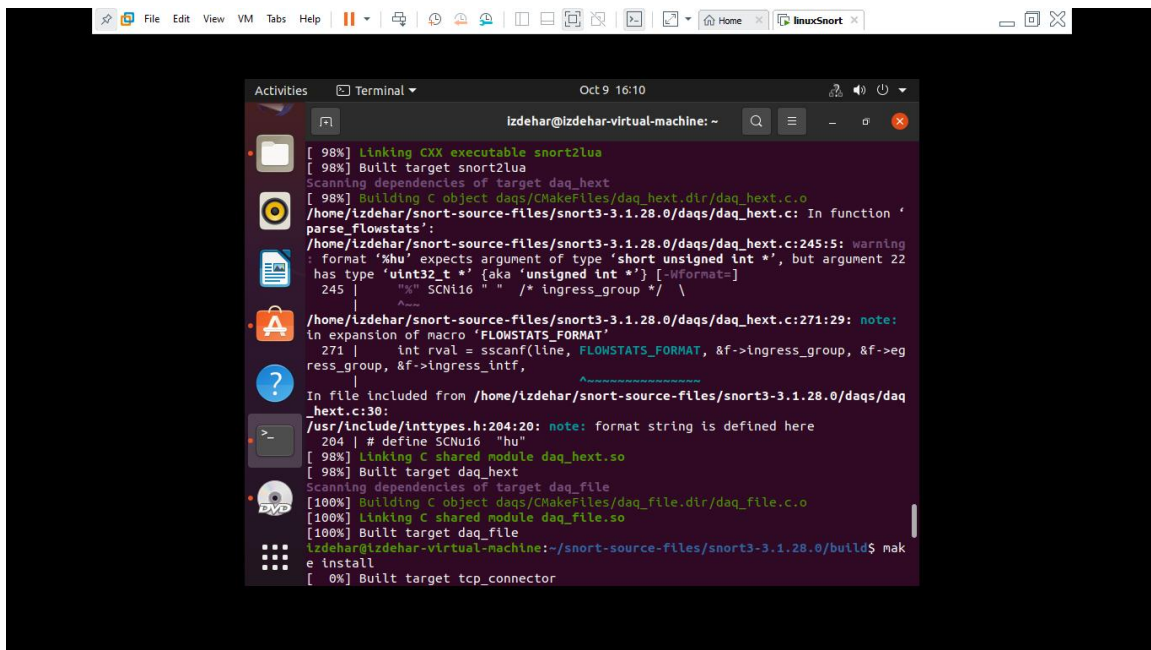
```
Oct 9 16:04
izdehar@izdehar-virtual-machine: ~
UG
EXE_LDFLAGS:
MODULE_LDFLAGS:
Feature options:
DAQ Modules: Static (afpacket;bpf;dump;fst;gmlb;nfq;pcap;savefile;trace)
libatomic: System-provided
Flatbuffers: OFF
Hyperscan: OFF
ICONV: ON
Libunwind: ON
LZMA: ON
RPC DB: Built-in
SafeC: OFF
TCMalloc: OFF
JEMalloc: OFF
UUID: ON
-----
-- Configuring done
-- Generating done
-- Build files have been written to: /home/izdehar/snort-source-files/snort3-3.1.28.0/build
izdehar@izdehar-virtual-machine:~/snort-source-files/snort3-3.1.28.0$ cd build
izdehar@izdehar-virtual-machine:~/snort-source-files/snort3-3.1.28.0/build$ make
Scanning dependencies of target tcp_connector
[ 0%] Building CXX object src/connectors/tcp_connector/CMakeFiles/tcp_connector.dir/tcp_connector.cc.o
```

2.3.15 Build directory: **Virtual-machine:~/snort-source-files/ snort3-3.1.28.0\$ cd build**  
**OUTPUT:**

```
Oct 9 16:04
izdehar@izdehar-virtual-machine: ~
UG
EXE_LDFLAGS:
MODULE_LDFLAGS:
Feature options:
DAQ Modules: Static (afpacket;bpf;dump;fst;gmlb;nfq;pcap;savefile;trace)
libatomic: System-provided
Flatbuffers: OFF
Hyperscan: OFF
ICONV: ON
Libunwind: ON
LZMA: ON
RPC DB: Built-in
SafeC: OFF
TCMalloc: OFF
JEMalloc: OFF
UUID: ON
-----
-- Configuring done
-- Generating done
-- Build files have been written to: /home/izdehar/snort-source-files/snort3-3.1.28.0/build
izdehar@izdehar-virtual-machine:~/snort-source-files/snort3-3.1.28.0$ cd build
izdehar@izdehar-virtual-machine:~/snort-source-files/snort3-3.1.28.0/build$ make
Scanning dependencies of target tcp_connector
[ 0%] Building CXX object src/connectors/tcp_connector/CMakeFiles/tcp_connector.dir/tcp_connector.cc.o
```

2.3.16 Make: **Virtual-machine:~/snort-source-files/ snort3-3.1.28.0/build\$ make**

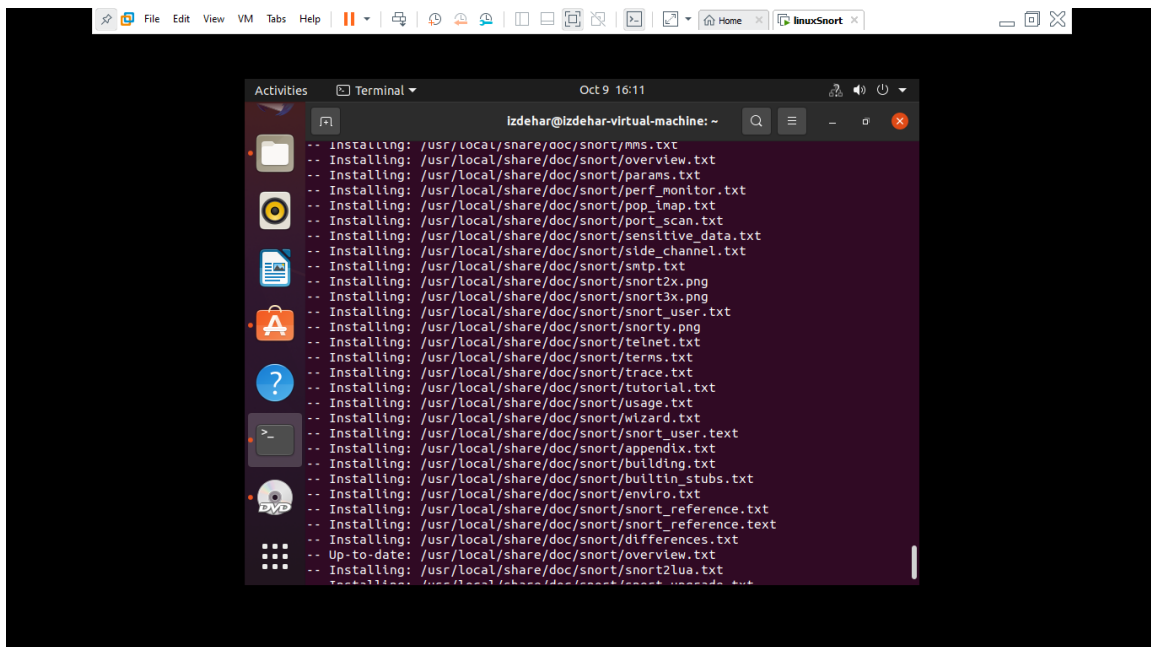
## OUTPUT:



```
izdehar@izdehar-virtual-machine: ~  
[ 98%] Linking CXX executable snort2lua  
[ 98%] Built target snort2lua  
Scanning dependencies of target daq_hex  
[ 98%] Building C object daqs/CMakeFiles/daq_hex.dir/daq_hex.c.o  
/home/izdehar/snort-source-files/snort3-3.1.28.0/daqs/daq_hex.c: In function 'parse_flowstats':  
/home/izdehar/snort-source-files/snort3-3.1.28.0/daqs/daq_hex.c:245:5: warning: format '%hu' expects argument of type 'short unsigned int *', but argument 22 has type 'uint32_t *' {aka 'unsigned int *'} [-Wformat=]  
245 |         "SCNU16" /* ingress_group */  
|         ^~~~~~  
/home/izdehar/snort-source-files/snort3-3.1.28.0/daqs/daq_hex.c:271:29: note: in expansion of macro 'FLOWSTATS_FORMAT'  
271 |         int rval = sscanf(line, FLOWSTATS_FORMAT, &f->ingress_group, &f->egress_group, &f->ingress_intf,  
|                               ^~~~~~  
In file included from /home/izdehar/snort-source-files/snort3-3.1.28.0/daqs/daq_hex.c:30:  
/usr/include/inttypes.h:204:20: note: format string is defined here  
204 | # define SCNU16 "hu"  
[ 98%] Linking C shared module daq_hex.so  
[ 98%] Built target daq_hex  
Scanning dependencies of target daq_file  
[100%] Building C object daqs/CMakeFiles/daq_file.dir/daq_file.c.o  
[100%] Linking C shared module daq_file.so  
[100%] Built target daq_file  
izdehar@izdehar-virtual-machine:~/snort-source-files/snort3-3.1.28.0/build$ make install  
[ 0%] Built target tcp_connector
```

### 2.3.17 Make install: **Virtual-machine:~/snort-source-files/ snort3-3.1.28.0/build\$** make install

## OUTPUT:



```
izdehar@izdehar-virtual-machine: ~  
-- Installing: /usr/local/share/doc/snort/mms.txt  
-- Installing: /usr/local/share/doc/snort/overview.txt  
-- Installing: /usr/local/share/doc/snort/params.txt  
-- Installing: /usr/local/share/doc/snort/perf_monitor.txt  
-- Installing: /usr/local/share/doc/snort/pop_inap.txt  
-- Installing: /usr/local/share/doc/snort/port_scan.txt  
-- Installing: /usr/local/share/doc/snort/sensitive_data.txt  
-- Installing: /usr/local/share/doc/snort/side_channel.txt  
-- Installing: /usr/local/share/doc/snort/smtp.txt  
-- Installing: /usr/local/share/doc/snort/snort2x.png  
-- Installing: /usr/local/share/doc/snort/snort3x.png  
-- Installing: /usr/local/share/doc/snort/snort_user.txt  
-- Installing: /usr/local/share/doc/snort/snort2y.png  
-- Installing: /usr/local/share/doc/snort/telnet.txt  
-- Installing: /usr/local/share/doc/snort/terms.txt  
-- Installing: /usr/local/share/doc/snort/trace.txt  
-- Installing: /usr/local/share/doc/snort/tutorial.txt  
-- Installing: /usr/local/share/doc/snort/usage.txt  
-- Installing: /usr/local/share/doc/snort/wizard.txt  
-- Installing: /usr/local/share/doc/snort/snort_user.text  
-- Installing: /usr/local/share/doc/snort/appendix.txt  
-- Installing: /usr/local/share/doc/snort/building.txt  
-- Installing: /usr/local/share/doc/snort/built_in_stubs.txt  
-- Installing: /usr/local/share/doc/snort/enviro.txt  
-- Installing: /usr/local/share/doc/snort/snort_reference.txt  
-- Installing: /usr/local/share/doc/snort/snort_reference.text  
-- Installing: /usr/local/share/doc/snort/differences.txt  
-- Up-to-date: /usr/local/share/doc/snort/overview.txt  
-- Installing: /usr/local/share/doc/snort/snort2lua.txt
```

### 2.3.18 Verify Snort 3 Installation: **Virtual-machine:\$** snort -v

## OUTPUT:

```
lzdehar@lzdehar-virtual-machine:~$ snort -v
-----
o")~  Snort++ 3.1.28.0
-----
Network Policy : policy id 0 :
-----
Inspection Policy : policy id 0 :
-----
pcap DAQ configured to passive.
-----
host_cache
    memcap: 8388608 bytes

Snort successfully validated the configuration (with 0 warnings).
o")~  Snort exiting
```

## REFERENCES

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6. 7 Ways to Get the Most from Your IDS/IPS. (April 2019). DarkReading.